Week 4, Day 3: Feature Engineering, Cross Validation & Model Comparison

.

1. Engineered Feature List

Feature	Type	Rule	Justification	MI Score
age_bucket	Categorical (binned)	pd.cut(age, fixed bins)	Income rises then plateaus with age, non-monotonic	0.0728
hours_bucket	Categorical (binned)	pd.cut(hours-per-week, fixed bins)	Part time / standard / overtime likely pay differently	0.0449
has_capital_gain	Binary flag	capital-gain > 0	Capital gain acts as a threshold signal, not continuous	0.0293
has_capital_loss	Binary flag	capital-loss > 0	Any recorded loss may signal investment activity	0.0076
higher_education	Binary flag	education-num >= 13	Bachelor's or higher is a natural, interpretable cutoff	0.0479
log_capital_gain	Numeric transform	log1p(capital-gain)	Tames right skew for tree models specifically	0.0805
edu_hours_interaction	Interaction	education-num * hours-per-week	Captures compounding effect of education and hours	0.0835
net_capital	Numeric	capital-gain - capital-loss	Nets gains and losses into one investment signal	0.1180

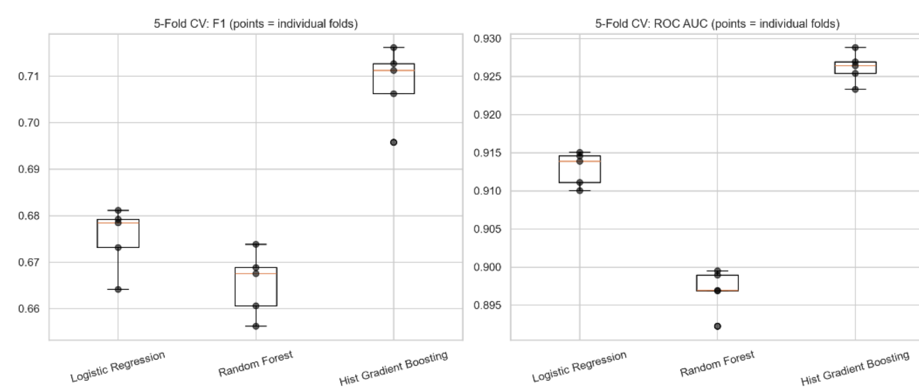
MI score = mutual_info_classif against the target (age_bucket and hours_bucket scores are summed across their one-hot dummy columns). net_capital carries the strongest univariate signal of any engineered feature, ahead even of the raw capital-gain/capital-loss columns individually, confirming that netting gains against losses captures more predictive information than either alone. has_capital_loss is the weakest, likely because so few rows have any recorded loss at all, limiting how much it can distinguish.

2. Cross Validated Model Comparison

A FunctionTransformer wrapping the feature engineering step feeds a three-branch ColumnTransformer (numeric, binary, categorical), refit inside every StratifiedKFold(k=5) split so no fold's validation rows influence another fold's training statistics. Three models were compared with identical preprocessing.

Model	F1 (mean +/- std)	ROC AUC (mean +/- std)	Fit time
Logistic Regression	0.6752 +/- 0.0061	0.9129 +/- 0.0020	3.2s
Random Forest	0.6654 +/- 0.0063	0.8969 +/- 0.0026	12.1s
Hist Gradient Boosting	0.7084 +/- 0.0071	0.9262 +/- 0.0018	4.8s

Hist Gradient Boosting is the clear top performer on both metrics while also being the second fastest to fit. Random Forest is both the weakest performer and the slowest to train, likely due to its 300 unpruned trees overfitting individual folds slightly, unlike Hist Gradient Boosting's built in regularization. All three models show low fold to fold variance, so these means are stable estimates.



3. Statistical Test Results

Comparison	Mean F1 diff	Paired t-test	Wilcoxon
Hist GB vs. Logistic Regression	0.0332	t = 14.044, p = 0.0001	W = 0.000, p = 0.0625

Hist Gradient Boosting beats Logistic Regression by a mean F1 gap of 0.0332, well above the 0.01 threshold for a meaningful difference. The paired t-test confirms this is significant (p = 0.0001). The Wilcoxon p-value of 0.0625 does not clear 0.05, but at n=5 folds this is the lowest p-value the exact test can ever report, even for a perfect result; W = 0.000 means Hist Gradient Boosting won all 5 of 5 folds with no exceptions. A large mean gap, a significant t-test, and a perfectly consistent per-fold win together indicate a genuine, practically meaningful difference, not a coincidence of one favorable split.

Feature importance and coefficients (engineered features only)

Feature	RF Importance	LogReg Coefficient
edu_hours_interaction	0.0960	-0.2030
net_capital	0.0537	2.4833
log_capital_gain	0.0413	-0.1542
higher_education	0.0291	0.0482
has_capital_gain	0.0117	-1.7902
has_capital_loss	0.0056	-2.2336
age_bucket (sum of dummies)	0.0459	range: -0.95 to 0.46
hours_bucket (sum of dummies)	0.0246	range: -0.86 to 0.33

edu_hours_interaction is Random Forest's single most important engineered feature (0.096), nearly double the next highest, consistent with trees being well suited to exploit a genuine multiplicative interaction that a single linear coefficient cannot represent directly. Logistic Regression, in contrast, assigns it a small negative coefficient, and shows a similar apparent contradiction for net_capital (strongly positive, 2.48) against has_capital_gain and has_capital_loss (both strongly negative). This is not evidence the features mean something different to each model, it reflects two compounding statistical effects rather than one. First, multicollinearity: net_capital, log_capital_gain, has_capital_gain, and has_capital_loss are all derived from the same underlying capital-gain/capital-loss columns, and edu_hours_interaction is a product of two features (education-num, hours-per-week) that remain in the model separately, so correlated predictors cause Logistic Regression to distribute credit for shared signal in ways that can flip individual signs while leaving overall predictions accurate. Second, Random Forest's default importance metric (mean decrease in impurity) is known to favor continuous, high-cardinality features over binary flags, which independently explains why has_capital_gain and has_capital_loss score low in the RF ranking even though they carry real, interpretable signal, this is a property of the importance metric itself, not evidence the flags are uninformative. Both models agree, despite these differences, that prime working-age brackets (36-55) and longer hours worked (41+) carry a positive, directionally sensible signal.

4. Feature Selection / Dimensionality Check

Configuration	F1 (mean +/- std)	ROC AUC (mean)	Fit time
Full feature set	0.7084 +/- 0.0071	0.9262	6.7s
SelectKBest (k=40)	0.6963 +/- 0.0067	0.9245	18.3s
SelectKBest (k=65)	0.7060 +/- 0.0060	0.9256	15.4s

SelectKBest (mutual information) was tested at two values of k. At k=40, F1 drops 0.0121 versus the full set, more than the 0.01 threshold for acceptance. At k=65, the gap narrows to 0.0024, confirming k=40 discarded genuinely informative features, particularly edu_hours_interaction. Neither k improves fit time; both are slower than the full set, since mutual_info_classif's KNN based estimator is expensive and must be refit inside every CV fold regardless of how many features are ultimately kept. Given Hist Gradient Boosting was already fast on the full set, selection never had a time advantage to offer.

5. Recommendation for Day 4 Tuning

- Recommended model: Hist Gradient Boosting, the clear top performer on mean F1 and ROC AUC, with a consistent 5-of-5 fold win over Logistic Regression.
- Recommended feature set: the full 20-feature set (5 raw numeric, 7 raw categorical, 8 engineered), no reduction from SelectKBest, since it did not clear the bar on performance or speed at either k tested.